EEL4745C: RTOS Final Project

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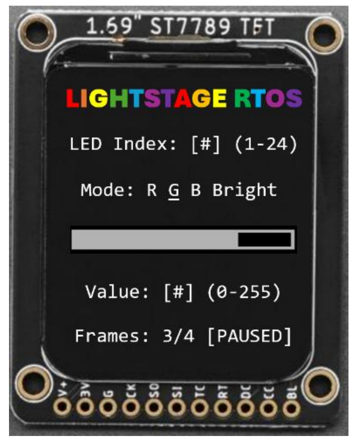
LightStage RTOS - One-Screen LED Programmer

Project Overview

LightStage RTOS is a real-time LED control system built on the TM4C123 LaunchPad and Multimod daughterboard. The system functions as a compact lighting-show console using the PCA9956B 24-channel LED driver. Users program LED colors (R, G, B) and brightness through a joystick and two buttons while a responsive LCD interface provides live feedback.

Major Components

- TM4C123GH6PM LaunchPad running the custom G8RTOS
- Multimod daughterboard with PCA9956B LED driver (I^2C bus A)
- ST7789 LCD display (SPI): single-screen UI showing LED index, mode, and value bar
- User input: Joystick [Left/Right = Next LED, Up/Down = Change Value, Press = Change Mode], SW1 [Save Frame], SW2 [Play/Pause Toggle]



RTOS Components

[Threads]

- **Aperiodic Threads**
  - **Joystick press:** cycle mode (R > G > B > Bright)
  - **SW1 press:** save current LED frame (up to 4)
  - **SW2 press:** toggle play/pause for frame playback
- **Periodic Threads**
  - **Sample joystick X/Y** (every 100 ms): store in FIFO
  - **Play saved frames** (when active): copy next frame & signal LED update
- **Background Threads**
  - **Input Thread (aggregator):** reads joystick FIFO and semaphores, updates LED index, color, and brightness, handles save/play commands, signals commit_sem.
  - **LED Commit Thread:** waits on commit_sem, locks I^2C, writes scaled RGB values to PCA9956B, releases bus.
  - **UI Render Thread:** redraws LCD (LED index, mode, value bar, frame status) using lcd_sem.

Semaphores (Purposes)

- I^2C transfer semaphore: exclusive PCA9956B bus access.
- LCD transfer semaphore: protect SPI LCD writes.
- Commit semaphore: signals LED update when data changes.
- SW1 press semaphore: frame save trigger.
- SW2 press semaphore: play/pause toggle.
- Joystick press semaphore: cycle color mode (R/G/B/Bright).

Peripherals (Purposes)

- I²C1: control PCA9956B PWM (24 LEDs).
- ADC channels: read joystick X/Y inputs.
- GPIO pins: detect SW1, SW2, and joystick press events.
- SPI: drive LCD display for UI feedback.
- UART: optional debug console.
- SysTick/NVIC: provide RTOS timing and interrupt scheduling.

Memory Management (IPC)

- Joystick FIFO: holds periodic X/Y samples for Input thread.
- pixels[24][3]: live RGB values (0-255) for each LED.
- frames[4][24][3]: stores up to four saved lighting frames.
- masterBright: global brightness scale (0-255).
- Flags: isPlaying, frameCount, playIdx track playback status.

Important Data Structures & Workflow

UI State Structure: tracks LED index, active channel (R/G/B/Bright), and adjustment step.

Workflow:

1. Joystick thread samples and writes to FIFO.
2. Button ISRs signal semaphores (SW1, SW2, press).
3. Input thread reads FIFO + signals, updates state, and posts commit_sem.
4. LED Commit thread writes new PWM values via I²C.
5. UI Render thread updates LCD.
6. Frame Player thread cycles through frames when playing.

Project Milestones

Milestone 1 - SF Demo: Core Input and LCD Interface

- Joystick controls LED index and value; press cycles color mode. LCD shows LED index, mode, and value bar. Demonstrates periodic + aperiodic threads, FIFO communication, and LCD mutex.

Milestone 2 - LED Control and Frame Storage

- I²C communication with PCA9956B working. LED Commit thread writes RGB/brightness to LEDs. SW1 saves current frame; LCD confirms 'Frame Saved.' Demonstrates mutexed I²C, commit signaling, and button interrupts.

Milestone 3 - Final Demo: Full LightStage RTOS (Playback + UI + LEDs)

- SW1 stores up to four frames; SW2 plays/pauses looped playback (~100 ms rate). LCD displays frame count and play/pause status. Demonstrates full RTOS integration periodic + aperiodic threads, FIFO/semaphore IPC, and multitasking across input, display, and LED control.